

Curriculum Vitae of Jayde Willingham

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EDUCATION

Doctor of Philosophy, Astrophysics

Swinburne University of Technology, Melbourne Australia

February 2026 - July 2029

Topic: Decoding the Light of the First Galaxies and their Supermassive Black Holes

Supervisor: Dr Darren Croton

Master of Research, Physics and Astronomy

Macquarie University, Sydney Australia

February 2025 - November 2025

Topic: Refining Dust Extinction Corrections for $H\alpha$ Luminosity Functions using radio-based calibration

Supervisor: Dr Andrew Hopkins

Bachelor of Philosophy, Physics and Astronomy

Macquarie University, Sydney Australia

February 2024 - November 2024

Completion with Distinction

Bachelor of Mathematics, Pure and Applied Mathematics

University of Newcastle, Newcastle Australia

February 2020 - November 2023

Completion with Distinction

Bachelor of Science, Physics

University of Newcastle, Newcastle Australia

February 2020 - November 2023

Completion with Distinction

PUBLICATIONS

J. Willingham, et al., “*EMU/GAMA: Refining Dust Extinction Corrections for $H\alpha$ Luminosity Functions Using Radio-Based Calibration.*”, (submitted; 2025).

K. Dage, et al., “*The Secret lives of open clusters.*”, DOI:10.1017/pasa.2026.10150 (2026)

J. Prathap, et al., “*EMU/GAMA: A statistical perspective on active galactic nuclei diagnostics.*”, (accepted; 2026).

J. Prathap, et al., “*EMU/GAMA: A new approach to characterising radio luminosity functions.*”, arXiv:2505.11453 (2025).

RESEARCH EXPERIENCE

ESO Summer Research Programme

Supervisor: Martyna Chruslinska and Mirko Curti

June 2025 - August 2025

- Over this 6-week project, I contributed to investigating the growth of dense stellar systems throughout cosmic time by developing and testing semi-empirical models of cluster evolution and their influence on the cosmic star formation history. One of seven chosen out of 484 applicants.

CSIRO Studentship

Supervisor: Dr Maxim Voronkov

November 2024 - February 2025

- We investigated the hyperfine structures of methanol masers in observational radio data. Our work included simulating these structures, modeling the data, and applying statistical analyses to better understand these poorly characterized features and their potential implications.

Bachelor of Philosophy Research

Supervisor: Professor Andrew Hopkins 2024

- This study quantifies uncertainties, evaluates the robustness of star formation rate (SFR) calculations, and compares their performance to model-derived SFRs, highlighting areas for improvement within the field.

Bachelor of Philosophy Research

Supervisor: Dr Tayyaba Zafar 2024

- We explored the viability of using observational radio fluxes from star-forming galaxies to constrain spectral energy distribution models, enabling improved parameter estimation.

Undergraduate Research

Supervisor: Associate Professor Lachlan Rogers 2022

- Here we investigate how the laser power dependence affects the photon auto-correlation time scales in the $g^{(2)}(\tau)$ function in order to make steps towards creating a rigorous model for analysing NV centres.

AWARDS

JADC Travel Grant, Swinburne University 2025
Research Poster Excellence Award, Australian Institute of Physics 2025
Research Excellence Scholarship, Macquarie University Jan - Nov 2025
High Achiever - Research Communications, Macquarie University 2024
Summer Research Scholarship, University of Newcastle Dec 2022
College Commendation List, University of Newcastle 2020 & 2021

OUTREACH AND COMMU- NICATION

Science Communication Writer, Astrobits.org 2026 - Ongoing
Astronomy Open Night Talk, Macquarie University 2025
The Quantum Experience Presenter 2025
Children's University Educator 2023 & 2024
ExperimentFest Educator 2024
HunterWISE Women in STEM Mentor 2022
UoN Undergraduate Project Colloquium, University of Newcastle October 2022
MQ BPhil Project Conference, Macquarie University May & October 2024

TEACHING EXPERIENCE

Casual Academic 2023 - present
PHYS1205 - Fundamentals of Engineering Physics
PHYS1210 - Advanced Physics I
PHYS1200 - Introductory Physics for the Life Sciences
STAT2110 - Engineering Statistics
School of Information and Physical Sciences, University of Newcastle
PHYS1510 - Engineering Physics
School of Mathematical and Physical Sciences, Macquarie University

Course Development 2025 - present
EPPHYS152 - Physics Essentials
EPPREP970 - Physics for Medical Radiation and Health Sciences *School of Information and Physical Sciences, University of Newcastle*

TECHNICAL SKILLS

Languages : Python, R, MATLAB, Mathematica, C.
General : Proficient in Microsoft/Google suites, Spyder and Jupyter notebooks, L^AT_EX, github, report writing, audio/visual editing, graphic design, Windows Subsystem for Linux.